

Supplementary Material

Simulation-Based Conformal Inference

What we do. We train an amortised predictor on synthetic trajectories from a known simulator, calibrate conformal prediction regions on a held-out synthetic set (optionally reweighted for covariate shift), and evaluate the regions on realistic or real observations whose labels are used only for scoring. Where needed (radial velocity), a label-free refinement maps the feed-forward estimate to a better fit before scoring. The forced-pendulum appendix checks the pipeline with known hidden labels; the radial-velocity appendix applies it to NASA archive curves.

1 Forced-Pendulum Synthetic Experiment: Full Detail

All results in this appendix come from the forced-pendulum run summarized in the main text. The purpose is to check the full information pattern where hidden labels are known for evaluation: the estimator is trained and calibrated on labelled simulations, while labels of the held-out “realistic” trajectories are unused until the final coverage calculation.

1.1 Setup

Dynamics. Starting from the nonlinear pendulum equation, we add a hidden nonlinear forcing term

$$\ddot{y}(t) = -\theta \sin y(t) + \lambda \cos(1.7t)y(t)\{1 - y(t)^2\}, \quad (1)$$

where $\theta = g/\ell$ and λ is an unknown forcing coefficient. The target is therefore the two-dimensional parameter $\vartheta = (\theta, \lambda)$. We use $T = 12$, $\Delta t = 0.1$, $y(0) = 0.8$, $\dot{y}(0) = 0$, and a velocity-Verlet integrator throughout. Parameters are sampled from $\theta \sim \text{Unif}(2, 16)$ and $\lambda \sim \text{Unif}(-2, 2)$.

Synthetic and realistic trajectories. Training and calibration trajectories are generated from (1) with independent Gaussian observation noise of variance $\epsilon = 0.006$. Realistic trajectories use the same ideal dynamics but include an additional unmodelled state perturbation. At each time step, with probability concentrated near the middle of the trajectory,

$$y(t) \leftarrow y(t) + \kappa y(t)\{1 - y(t)^2\},$$

where $\kappa \sim \text{Unif}(0.005, 0.035)$. This perturbation is not included in the simulator used by either conformal calibration or the Bayesian likelihood. It plays the same role, in miniature, as unmodelled physical structure in the radial-velocity experiment.

Predictor and conformal region. The encoder ψ is a distance-weighted k -nearest-neighbour regressor with $k = 5$, trained on 1200 labelled synthetic trajectories. Each trajectory is represented by Fourier features and a small set of time-domain summaries. A separate calibration set

of 400 synthetic trajectories is used for split conformal prediction, with coordinate-wise score

$$s_j(y, \vartheta) = |\psi_j(y) - \vartheta_j|, \quad j \in \{\theta, \lambda\}.$$

The target joint confidence is 0.68, matching the Bayesian one-standard-deviation comparison. Since the conformal region is rectangular, we use the Bonferroni coordinate confidence

$$1 - \alpha_j = 1 - \frac{1 - 0.68}{2} = 0.84.$$

The resulting box is

$$\Gamma(y) = [\psi_\theta(y) - q_\theta, \psi_\theta(y) + q_\theta] \times [\psi_\lambda(y) - q_\lambda, \psi_\lambda(y) + q_\lambda],$$

with the lower endpoint of θ clipped to remain positive. In the reported run, $q_\theta = 0.0375$ and $q_\lambda = 0.7506$.

Bayesian baseline. For each held-out trajectory we also compute a Bayesian grid posterior over (θ, λ) . The likelihood uses the ideal simulator h from (1) and independent Gaussian observation noise:

$$p(y \mid \theta, \lambda) \propto \exp \left\{ -\frac{1}{2\epsilon} \sum_t (y_t - h_t(\theta, \lambda))^2 \right\}.$$

The prior is uniform on $[1.5, 18] \times [-2.5, 2.5]$, and the posterior is evaluated on a 70×61 grid. We report the posterior mean and coordinate-wise mean $\pm$ one posterior standard deviation. This is a simple feasible Bayesian comparator for the synthetic experiment, not a claim about the best possible Bayesian method for this model.

1.2 Results

Parameter identification on the 45 held-out realistic trajectories is shown in Figure 1 of the main paper (conformal intervals against Bayesian one-standard-deviation intervals). Table 1 summarises the same run numerically.

$\theta = g/\ell$ is easy because it controls the dominant oscillation rate, while λ is the informative direction. The conformal point estimator has lower MSE on both coordinates. More importantly, the conformal box contains the full hidden pair (θ, λ) for 35/45 held-out trajectories (joint coverage 0.778 at nominal 0.68). The Bayesian

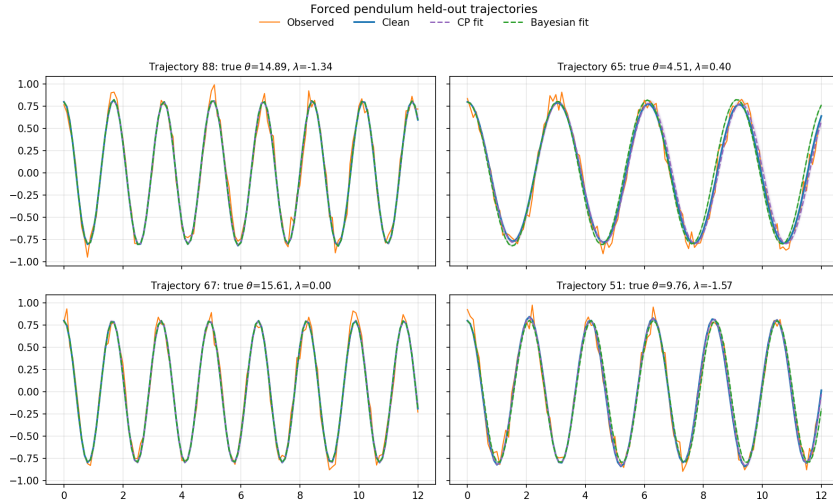


Figure 1: Trajectory-level view for four held-out realistic pendula. Each panel shows the noisy observations, the clean generating trajectory, and clean simulator trajectories from the CP and Bayesian point estimates. Faint curves are sampled from the conformal box and from the Bayesian one-standard-deviation rectangle.

Metric	CP	Bayesian grid
MSE, $\theta = g/\ell$	4.7×10^{-4}	5.2×10^{-3}
MSE, λ	0.222	0.486
Coverage, θ	0.867	0.044
Coverage, λ	0.844	0.244
Joint coverage	0.778	0.044

Table 1: Held-out forced-pendulum results. CP is calibrated to joint target confidence 0.68; the Bayesian columns use coordinate-wise one-standard-deviation intervals.

one-standard-deviation rectangle contains the full pair for 2/45. Narrow Bayesian intervals in θ drive the low joint coverage.

Figure 1 shows why this is an uncertainty-estimation example rather than merely a curve-fitting example. Fitted clean simulators can track the observed trajectory even when the latent forcing parameter is poorly covered. A visually plausible curve therefore does not imply reliable uncertainty in the hidden parameter — the same qualitative issue that motivates the radial-velocity experiment.

Limitations. The experiment is deliberately small and fully labelled: a pipeline check rather than evidence about pendulum physics. The Bayesian baseline is intentionally simple (70×61 grid). The point is narrower: with the same approximate simulator and finite held-out data, conformal calibration gives a conservative prediction region in the latent physical parameter space.

2 Radial-Velocity Experiment: Full Detail

All results in this appendix come from a single calibration run unless stated otherwise: refinement uses $K = 200$ gradient steps from $M = 5$ restarts with the semi-amplitude refitted analytically, $n_{\text{cal}} = 400$, and the test set is 34 independent host stars.

2.1 Setup

Data. We use radial-velocity time series from the NASA Exoplanet Archive together with tabulated orbital elements from its Planetary Systems table. The corpus comprises 1,071 RV curves across 411 host stars (1,469 star–planet rows), measured by ground-based spectrographs — predominantly Keck/HIRES, the Lick 3.0 m Shane, the Anglo-Australian Telescope and HARPS. The median curve has 30 observations over a 1,336 d baseline. Restricting to single-planet systems and splitting *by host* gives 292 training, 78 validation and 74 test series. Splitting by host rather than by curve matters: several hosts appear with multiple independent campaigns, and a curve-level split would leak the same system across train and test.

Two further cuts apply at test time. Requiring $n_{\text{obs}} \geq 10$ leaves 57 test series. We then require at least five distinct observing *nights*: a curve whose observations all fall inside a single night cannot constrain a period of hundreds of days.¹ This filter is deliberately *label-free* — it reads only the observing pattern — and removes 6 of the 57. A label-aware version (baseline $\geq$ one tabulated

¹HD 17156 is the motivating case: 13 points inside a 0.102 d window, a Rossiter–McLaughlin sequence, against a tabulated period of 21.2 d.

period) removes the same six.

The corpus stores one row per RV file, so the 51 surviving series come from only 34 distinct hosts (HD 179949 contributes four; 51 Peg three). We keep the best-sampled series per host and report coverage over **34 independent systems**. Section 2.5 reports all three populations side by side.

Target parameters. Here $d = 3$, with $\theta = (\log_{10} P, \log_{10} K, e)$. These are the parameters identifiable from radial velocity alone. The argument of periastron ω is estimated by the model but is *not reported*: its conformal region is vacuous. On the held-out synthetic split ($n = 2000$) the point predictor’s mean absolute error in ω is 95.2° (against 90° from a uniform guess), and R^2 on $\cos \omega$ and $\sin \omega$ is -1.43 and -1.45 . We exclude ω from reported coordinates and from the Bonferroni correction, but retain it inside the model because e is parameterised through $(e \cos \omega, e \sin \omega)$ and T_{peri} is refitted against it. T_{peri} and the systemic offset enter h linearly and are refitted analytically at each evaluation.

Simulator and priors. The noiseless integrator h solves Kepler’s equation on the observation’s own time grid. Marginal priors $p(\bar{\theta})$ are independent empirical histograms fitted on the training split only; eccentricity follows a Beta(0.867, 3.03) prior. The noise model is a sparse variational Gaussian process fitted to pooled training residuals $r = y - h(\theta)$. Observation cadences and uncertainties are bootstrapped from real training curves. Draws whose maximum deviation from the noiseless model exceeds a bound estimated on real training curves are rejected at generation; on this run the bound was 5.99 (in units of the curve’s RV standard deviation) and discarded 1,200 of 2,500 prior draws (48.0%).

Encoder. The feature map f in $\psi = g \circ f$ embeds a curve into $l = 74$ dimensions: a 64-dimensional spectral encoding plus 10 time-domain summaries. The spectral block is the low-frequency power spectrum of a smoothing-spline interpolant on a uniform grid; the summaries include the peak period and peak power of a generalised Lomb–Scargle periodogram (Zechmeister and Kürster, 2009). We retain spectral power only; absolute phase is discarded because T_{peri} is refitted analytically.

The regression head g is a three-layer perceptron, $74 \rightarrow 128 \rightarrow 64 \rightarrow 5$, with ReLU activations, trained for 200 epochs with AdamW at learning rate 10^{-3} , batch size 64, and early stopping on a 20% validation split with patience 30. Inputs and targets are standardised on the training split alone. Outputs are $(\log_{10} P, \log_{10} K, e, \cos \omega, \sin \omega)$; e is clipped to $[0, 0.99]$ at evaluation. Training uses synthetic data alone against $\bar{\theta}$; catalogue values are never a training target.

Why the refinement is necessary. The median observing baseline spans 21.7 orbits, so phase coherence

requires $\Delta P/P \lesssim 0.5\%$ (0.002 dex in $\log_{10} P$) — far beyond what a summary-feature regressor achieves (ours reaches $R^2 \approx 0.79$ against a 1.05 dex prior spread). We therefore report $\psi'(y) = \text{GD}_K(\psi(y))$ with $K = 200$ steps at learning rate 0.02 and $M = 5$ restarts from the strongest periodogram peaks, keeping the lowest-loss fit, with K refitted analytically. ψ' is still a deterministic map from the observation alone, requires no label, and is applied identically to calibration, tuning and test sets. The surrogate label θ^* is untouched by the refinement. Cost is quantified in Section 2.4.

Protocol. Calibration uses $n_{\text{cal}} = 400$ synthetic curves, with a further 100 for hyperparameter tuning and 400 held out as a synthetic test set. The real validation and test splits are never seen during calibration; γ is tuned on the real validation split (33 systems). The likelihood-ratio discriminator is fitted on a separate synthetic sample against the real training split, on the 74-dimensional summary features. Prediction regions use the Bonferroni level $1 - \alpha/d$ with $d = 3$. The tuned normalisation weight selects $b = 0.50$ for the surrogate strategy and $b = 0.75$ for the naive strategies. The tuned γ is 2.858×10^{-3} . The effective calibration sample size after reweighting is 262.6 of 400, with no weight clipped. A budget $\beta = 0.0099$ of the error level is spent on the surrogate gap, leaving $\alpha - \beta$ for the main quantile.

2.2 Validity

Coverage across error levels. Table 2 gives joint coverage at all seven calibrated levels. Figure 2 plots the same quantities against nominal.

In brief: scored against θ^* on synthetic data, coverage is 0.905 at nominal 0.90 — essentially exact, and slightly closer to nominal than the $\bar{\theta}$ figure of 0.915. On real data the procedure is conservative (34/34 at 0.90; 30/34 at 0.68), from Bonferroni and reweighting; that is an efficiency statement, not a validity failure. With only 34 hosts, one system moves the real figure by three points, so we rest no claim on comparing real-data coverage between variants and rely on the synthetic column ($n = 400$).

The $\Delta s = 0$ validity gap and Δ_c . Section 3 of the *main paper* proves a coverage guarantee after setting a surrogate-score gap Δs to zero for tractability. In words: the theorem is stated as if the conformity score at the surrogate label θ^* and the score that would appear in an ideal (fully observed) bound were the same. In reality those two scores can differ slightly on each curve. That per-curve difference is Δs .

Setting $\Delta s = 0$ therefore leaves a thin band of scores around the calibrated threshold that the proof does not account for. We call the mass of curves that fall in that ignored band the *validity gap*: it is the empirical price of the $\Delta s = 0$ simplification. What we estimate empirically, on the synthetic test set where both θ^* and $\bar{\theta}$ are known,

Nominal $1 - \alpha$	Synthetic ($n=400$)		Real ($n=34$)	
	vs $\bar{\theta}$	vs θ^*	unweighted	weighted
0.95	0.973	0.960	1.000	1.000
0.90	0.915	0.905	1.000	1.000
0.85	0.895	0.883	1.000	0.971
0.80	0.833	0.850	0.971	0.941
0.70	0.748	0.770	0.971	0.882
0.68	0.738	0.763	0.941	0.882
0.60	0.670	0.710	0.882	0.853

Table 2: Joint coverage of the surrogate strategy under the δ_c, δ_y normalisation, Bonferroni-corrected across $d = 3$. “vs $\bar{\theta}$ ” scores against the data-generating parameters; “vs θ^* ” against the surrogate label that the validity theorem bounds. No interval is unbounded. The θ^* column is at or closer to nominal than the $\bar{\theta}$ column at every level below 0.90.

Nominal	$\log_{10} P$	$\log_{10} K$	e
0.90	0.175	0.920	0.763
0.68	0.310	0.898	0.843

Table 3: Band mass of the $\Delta s = 0$ gap on the synthetic test set, scored against θ^* . The gap is small in $\log_{10} P$ and large in $\log_{10} K$ and e .

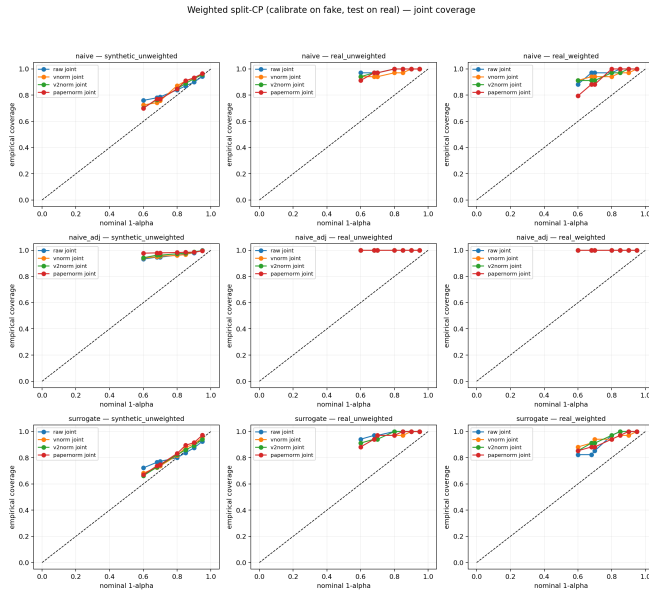


Figure 2: Empirical joint coverage against nominal $1 - \alpha$ for every calibration strategy (rows) and evaluation population (columns), with all four score normalisations overlaid. The dashed diagonal is exact calibration. The reported configuration is the red curve in the bottom-right panel.

is the *band mass* in Table 3 — the fraction of curves whose score lands inside that ignored band, per coordinate. A small band mass means the simplification is harmless for that coordinate; a large one means many curves sit in the unaccounted strip (as for $\log_{10} K$ and e below). Figure 3 shows the underlying Δs distribution on the calibration set.

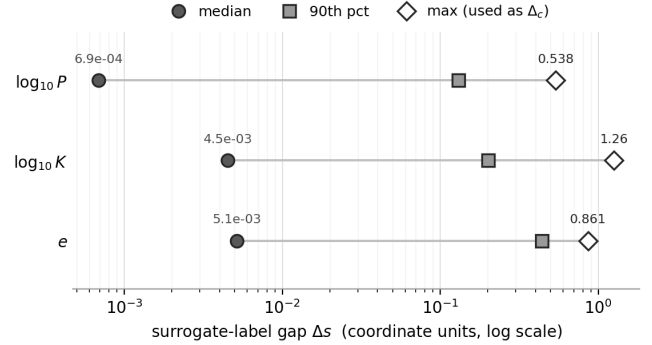


Figure 3: Distribution of the surrogate gap Δs over the calibration set, per reported coordinate.

The complementary quantity Δ_c is the conformal *surrogate-gap adjustment*

$$\Delta_c = \sup_y |s(y, \theta^*) - s(y, \bar{\theta})|,$$

a non-negative widening of the calibrated threshold that restores a valid bound when Δs is not identically zero. In the naive+ Δ_c rows of Table 4, Δ_c is taken as the maximum of the empirical Δs distribution (as the worst-case bound requires). Empirically Δ_c is small at the median and heavy-tailed (medians 6.9×10^{-4} , 4.5×10^{-3} , 5.1×10^{-3} dex for $\log_{10} P$, $\log_{10} K$, e ; maxima 0.538, 1.258, 0.861), so the max-based adjustment is driven by a few curves and produces the marked over-coverage of those rows.

Calibration strategies and score normalisations. Table 4 gives the full grid. Naive rows are ordinary split CP on $\bar{\theta}$ without surrogate labels or reweighting; they exist only on synthetic data and are a conventional baseline

analogue, not a competitor on real systems. **Closer to nominal is better** than higher coverage: over-coverage (e.g. naive+ Δ_c) is not preferred.

strategy	norm	nominal 0.90				nominal 0.68			
		syn. ($\bar{\theta}$)	syn. (θ^*)	real unw.	real wtd.	syn. ($\bar{\theta}$)	syn. (θ^*)	real unw.	real wtd.
naive ($\bar{\theta}$)	raw	0.900	0.930	1.000	1.000	0.780	0.825	0.971	0.971
	v_y	0.930	0.935	1.000	0.971	0.743	0.782	0.941	0.941
	$v_y + v_c$	0.925	0.938	1.000	1.000	0.765	0.797	0.971	0.912
	δ_c, δ_y	0.932	0.925	1.000	1.000	0.770	0.782	0.971	0.882
naive + Δ_c	raw	0.980	0.988	1.000	1.000	0.948	0.958	1.000	1.000
	v_y	0.985	0.988	1.000	1.000	0.950	0.958	1.000	1.000
	$v_y + v_c$	0.988	0.988	1.000	1.000	0.963	0.955	1.000	1.000
	δ_c, δ_y	0.985	0.985	1.000	1.000	0.980	0.983	1.000	1.000
surrogate (θ^*)	raw	0.875	0.910	1.000	1.000	0.765	0.815	0.971	0.824
	v_y	0.905	0.917	1.000	0.971	0.725	0.770	0.941	0.912
	$v_y + v_c$	0.892	0.902	1.000	1.000	0.728	0.775	0.941	0.912
	δ_c, δ_y	0.915	0.905	1.000	1.000	0.738	0.763	0.941	0.882

Table 4: Joint coverage at nominal 0.90 and 0.68. Bold is the **reported** configuration (surrogate with δ_c, δ_y), chosen for being among the closest to nominal on the synthetic θ^* columns — not for highest coverage. Naive+ Δ_c over-covers and is not preferred. No configuration produces an unbounded interval on any real system.

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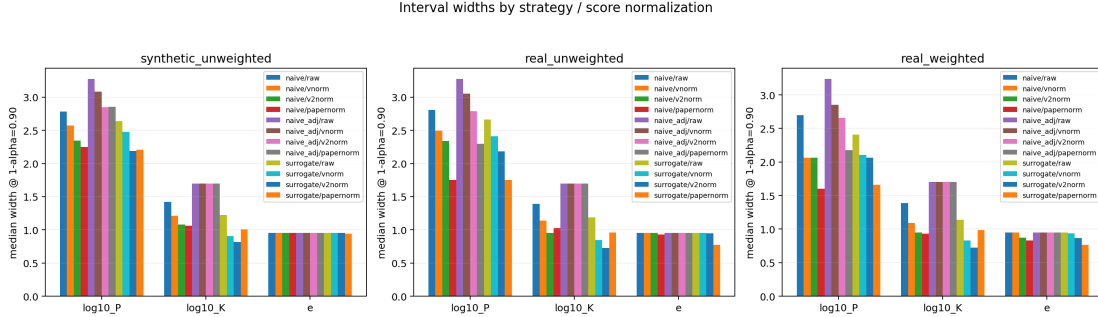


Figure 4: Distribution of per-system interval widths across calibrated error levels.

Union regions. Period aliases make a single interval spend measure on gaps between modes. The union merges per-restart intervals. Table 6 compares like-for-like total measure: box = $2 \times$ the half-width of Table ?? (unclipped); union = unclipped merged length.

The union is **not** uniformly tighter: it widens $\log_{10} P$ and e and shrinks $\log_{10} K$ modestly. Against θ^* on synthetic data, joint coverage improves (0.933 vs 0.905 at nominal 0.90); against $\bar{\theta}$ it falls slightly. On real data both shapes cover 34/34. The defensible claim is that the union trades measure for coverage of the surrogate label.

The Earth-like sample. Table 7 compares the ten most Earth-like held-out systems to catalogue 1σ uncertainties at nominal 0.68. Earth-likeness selects long periods, so regions widen — the adaptive normalisation reporting how little the data constrains. **Disclosures:** six test and four validation systems; γ was tuned on validation, so four rows informed their own normalisation. Coverage figures over the 34 test hosts are unaffected. Main paper Figure 2

Nominal, real weighted	box	union	change
0.90 $\log_{10} P$	2.200	2.527	+15%
$\log_{10} K$	1.242	1.135	−9%
e	1.170	1.392	+19%
0.68 $\log_{10} P$	0.915	1.016	+11%
$\log_{10} K$	0.544	0.491	−10%
e	0.810	1.087	+34%

Table 6: Total reported measure, single box against union (unclipped full widths). This supersedes the main-text union comparison.

shows the corresponding conformal regions (crosses mark $\psi'(y)$); we do not repeat that figure here. Figure 5 instead shows the observed RV trajectory for *each* of the ten Earth-

like systems, with catalogue and $\psi'(y)$ reconstructions overlaid.

Host	Split	Orbital period P (d)		Semi-amplitude K (m s ⁻¹)		Eccentricity e	
		Catalogue 1σ	CP 0.68	Catalogue 1σ	CP 0.68	Catalogue 1σ	CP 0.68
GJ 649	test	600 ± 1.7	[315, 2378]	9.71 ± 0.54	[8.77, 37.4]	0.08 ± 0.062	[0.00, 0.68]
HD 141004	val	15.5 ± 0.0017	[1.33, 97.0]	3.45 ± 0.36	[4.96, 67.3]	0.16 ± 0.11	[0.00, 0.99]
HD 2952	test	312 ± 1.5	[12.7, 738]	18.4 ± 2.5	[9.86, 95.9]	0.27 ± 0.16	[0.00, 0.99]
HD 147513	val	528 ± 6.3	[186, 1312]	29.3 ± 1.8	[22.1, 76.1]	0.26 ± 0.050	[0.00, 0.99]
HD 173416	test	322 ± 1.4	[8.03, 4163]	35.3 ± 4.3	[28.7, 132]	0.21 ± 0.092	[0.00, 0.99]
HD 10647	test	989 ± 8.1	[436, 1818]	18.1 ± 1.7	[12.1, 47.4]	0.15 ± 0.080	[0.27, 0.88]
HD 231701	val	142 ± 0.067	[34.7, 458]	39.2 ± 1.2	[17.7, 97.6]	0.13 ± 0.032	[0.00, 0.68]
HD 206610	test	673 ± 3.3	[321, 2261]	35.4 ± 1.0	[21.2, 72.9]	0.10 ± 0.042	[0.00, 0.88]
HD 32518	val	157 ± 0.090	[54.7, 224]	98.9 ± 5.7	[67.0, 143]	0.03 ± 0.027	[0.00, 0.50]
HD 73534	test	1750 ± 24	[417, 2936]	17.1 ± 0.93	[18.6, 64.1]	0.13 ± 0.057	[0.00, 0.87]

Table 7: Ten most Earth-like held-out systems (six test, four validation): catalogue $\pm$ against CP regions at nominal 0.68. Joint coverage is **7 of 10** (misses: HD 141004 and HD 73534 in K ; HD 10647 in e).

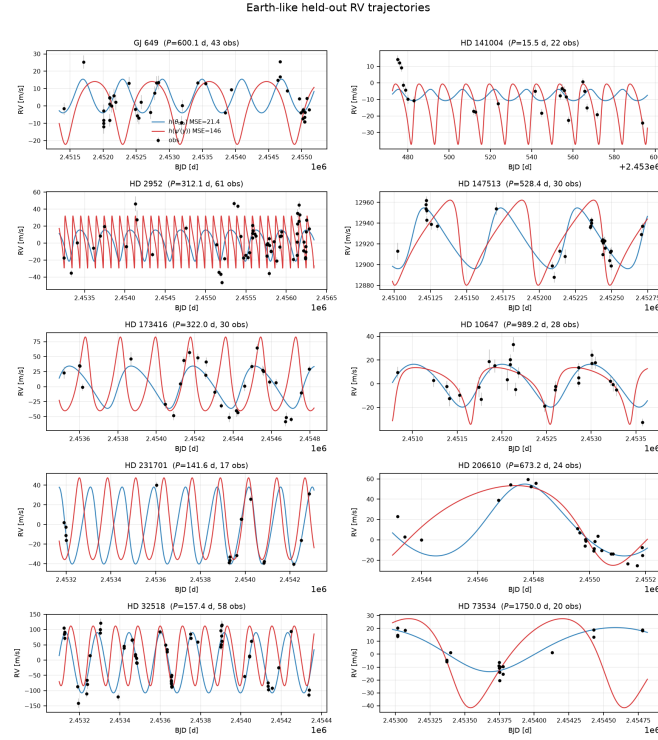


Figure 5: All ten Earth-like held-out systems from Table 7 (one panel per host): observations with measurement uncertainties, and noiseless Keplerian reconstructions at the catalogue parameters and at the refined prediction $\psi'(y)$.

Surrogate labels against the catalogue. On the 51 held-out test curves, median reconstruction MSE is 0.072 for θ^* and 0.090 for the tabulated parameters; θ^* is at or below the catalogue on all 51 curves. The refined predictor ψ' reaches 0.176 and beats or matches the catalogue on 51% of curves (2% for unrefined ψ). Main paper Figures 3–

4 show trajectory and MSE comparisons for a subset of systems; we do not repeat those figures here.

2.4 Computational cost

Stage	ms / system
Featurisation	0.018
ψ forward pass	0.028
ψ total	0.046
ψ' refinement ($K=200$, $M=5$)	55 913
θ^* gradient descent	9 903

Table 8: Per-system cost, 51 real curves, one thread, three repeats.

Every reported interval is centred on ψ' , which costs ~ 56 s per system — not the 0.046 ms amortised map alone. θ^* is cheaper by a factor of ~ 5.6 . We claim no speed advantage over per-target optimisation at this operating point; refinement buys accuracy (MSE 1.12 $\rightarrow$ 0.18; catalogue agreement 2% $\rightarrow$ 51%). Cost is dominated by $M \times K$ and is reducible, but none of the paper results use the faster Newton tolerance (that would require full recalibration).

2.5 Robustness and limitations

Test-set population. Table 9 repeats the analysis on the 34 hosts, 51 filtered series, and all 57 series with $n_{\text{obs}} \geq 10$.

	34 hosts	51 series	57 series
0.90 half-width $\log_{10} P$	1.100	1.211	1.284
$\log_{10} K$	0.621	0.594	0.633
e	0.585	0.601	0.639
0.68 half-width $\log_{10} P$	0.458	0.583	0.658
$\log_{10} K$	0.272	0.268	0.278
e	0.405	0.417	0.438
real weighted joint, 0.90	1.000	0.941	0.930
real weighted joint, 0.68	0.882	0.843	0.807
unbounded intervals	0	0	1.8%

Table 9: All three test populations at matched $K = 200$. Widths grow and coverage falls as harder series are restored; only the 57-series arm has an unbounded interval (one system).

Simulator misspecification. We add a correlated 30 d component (standing for stellar activity) at fraction f of each curve’s RV standard deviation on the test set only. Table 10: against θ^* , coverage holds at nominal through $f = 0.5$ and breaks at $f = 1.0$. The $\theta^* - \bar{\theta}$ margin widens with f , as predicted. There is no valid $f = 0$ baseline in this sweep (perturbation disables the basin rejection filter and changes the quantile). A multi-planet mean-structure probe was inconclusive by construction.

f	nominal 0.90		nominal 0.68	
	vs $\bar{\theta}$	vs θ^*	vs $\bar{\theta}$	vs θ^*
0.25	0.898	0.903	0.705	0.750
0.50	0.863	0.898	0.645	0.723
1.00	0.765	0.808	0.448	0.553

Table 10: Joint coverage under simulator misspecification ($n = 400$). Widths are identical across arms.

Is θ^* the global optimum? The implementation returns a multi-start gradient-descent limit, not a proven global minimiser. ψ' beats θ^* on reconstruction loss for 23% of synthetic curves (median loss ratio 1.16) and 7.8% of real curves (median ratio 1.59; parameter margins negligible). On real data the two are operationally close; the definition should read as the GD limit. The basin assumption is unaffected.

Assumption constants. On 25 prior draws, $\kappa(H)$ has median 1.62×10^4 and maximum 8.80×10^7 ; $\|\nabla h\|$ has median 172. Conditioning is benign at the median and severe in the tail — consistent with the width spread in Table ??.

The likelihood-ratio discriminator. Table 11: shift signal lives in dimensions 41–74; at $C = 1$ the fit spends 56% of ESS. $C = 0.03$ is better on every diagnostic. A full recalibration at $C = 0.03$ (Table 12) raises ESS from 263 to 337 and real coverage at 0.68 from 0.882 to 0.912, at the cost of 4–6% wider half-widths at 0.90. We report

$C = 1$ for consistency with the rest of the paper and flag $C = 0.03$ as better justified.

	AUC _{oof}	log-loss _{oof}	skill	ESS	$w_{\max}$
<i>by feature dimension, $C = 1$</i>					
$d = 10$	0.558	0.6545	+0.008	91.9%	2.06
$d = 40$	0.558	0.6546	+0.008	91.8%	2.07
$d = 74$	0.648	0.6403	+0.029	44.1%	18.7
<i>by regularisation, $d = 74$</i>					
$C = 0.01$	0.640	0.6371	+0.034	87.5%	3.75
$C = 0.03$	0.652	0.6312	+0.043	71.7%	8.07
$C = 0.10$	0.653	0.6332	+0.040	54.2%	14.2
$C = 1.00$	0.648	0.6403	+0.029	44.1%	18.7

Table 11: Discriminator diagnostics. Reported runs use $C = 1$ at $d = 74$.

	$C = 1$ (reported)	$C = 0.03$
Effective sample size	262.6	336.7
Weights clipped	0%	0%
Real joint, nominal 0.90	1.000	1.000
Real joint, nominal 0.68	0.882	0.912
Synthetic vs θ^* , 0.90	0.905	0.905
Synthetic vs $\bar{\theta}$, 0.90	0.915	0.915
0.90 half-width $\log_{10} P$	1.100	1.160
0.68 half-width $\log_{10} P$	0.458	0.614
0.68 half-width e	0.405	0.420

Table 12: Full recalibration at $C = 0.03$ vs reported $C = 1$.

References

M. Zechmeister and M. Kürster. The generalised Lomb–Scargle periodogram. *Astronomy & Astrophysics*, 496:577–584, 2009.